Graid Technology Inc.

SupremeRAID™ by Graid Product Introduction

MK Uang

Graid Technology Inc.

APAC General Manager

03/ 2025

GRAIDTECH.COM

SupremeRAID™

4th Revolution of GPGPU Application
Highest NVMe Data IO for Big Data protection

Graid Technology Inc.

- Graid GPU RAID in Big Data Era
- Phase in Timing
- Supreme100 Workstation
- SupremeStore
  HPC Cluster Storage
- PowerEdge R760 OEM with Graid Sol.
- Global Partners
- The Safest NVMe SSD GPU RAID Card
- Product roadmap update

SupremeStore/ Supreme100/ SupremeRAID™

Graid Technology Inc.

Graid GPU RAID
The Revolution in Big Data Era

Opportunity of Graid Solution

GPU RAID era in Big Data

As data volume grows, storage devices upgrade, and more emphasis on data security!

HDD

SSD

more than 20 times performance improvement

Traditional RAID

Hardware architecture with **serial data transmission** results the performance can not be fully unleashed.

Software RAID

Direct connection architecture relies on CPU computing, competing CPU resources and reducing system efficiency.

SDS

SDS/ EC

Relying on software and hardware resources, but still **CPU serial process**ing.

Graid GPU RAID

GPU-based parallel computing breaks traditional architecture **unleashing performance.**

The Best Solution For Performance and Data Protection in the Big Data Era

High Compatibility

Verified by Top One Tier Companies

Plug and Play, Easy to Use

SERVER

Please check the AVL for more information

SSD

High Compatibility

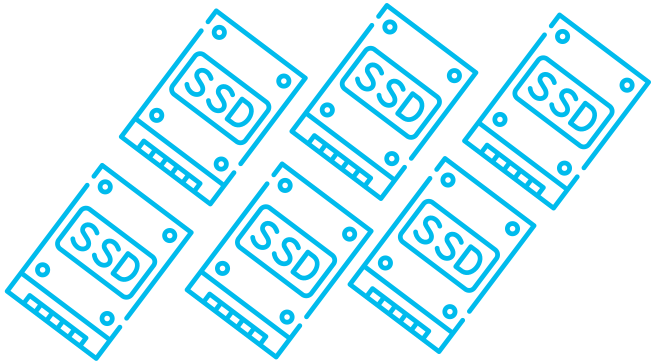
Verified by Top One Tier Companies
Plug and Play, Easy to Use

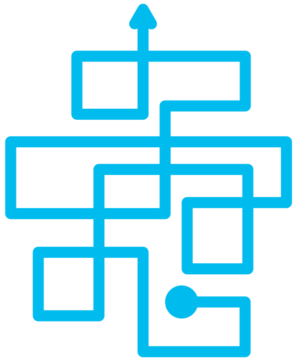
Linux Distribution	x86_64
AlmaLinux	8.5-8.10 (kernel 4.18), 9.0-9.5 (Kernel 5.14)
CentOS	7.9 (Kernel 3.10 and 4.18), 8.3, 8.4, 8.5 (Kernel 4.18)
Debian	11.6 (Kernel 5.10), 12 (Kernel 6.1)
openSUSE Leap	15.2-15.3 (Kernel 5.3), 15.4-15.5 (Kernel 5.14)
Oracle Linux	8.7-8.10 (RHCK 4.18 and UEK 5.15), 9.1-9.4 (RHCK 5.14 and UEK 5.15)
RHEL	7.9 (Kernel 3.10 or 4.18), 8.3-8.10 (Kernel 4.18), 9.0-9.5 (Kernel 5.14)
Proxmox VE	8.1 (Kernel 6.5), 8.2 (Kernel 6.8)
Rocky Linux	8.5-8.10 (Kernel 4.18), 9.0-9.5 (Kernel 5.14)
SLES	15 SP2-SP3 (Kernel 5.3), 15 SP4-SP5 (Kernel 5.14)
Ubuntu	20.04.0-20.04.4 (Kernel 5.4 and 5.15), 22.04.0-22.04.2 (Kernel 5.15), 22.04.3-22.04.4 (Kernel 6.2), 24.04 (Kernel 6.8)

Windows Version	Arch
Windows Server 2019	x86_64
Windows Server 2022	x86_64
Windows Server 2025	x86_64
Windows 11	x86_64

Phase in Timing _ SupremeRAID™

Server support more than 4 NVMe SSDs but Poor Performance & can’t build in one Raid group if more than 8 SSDs





Buying many SSDs, performance still falls short.

Even with 2-3 RAID cards, still can't form a RAID group?

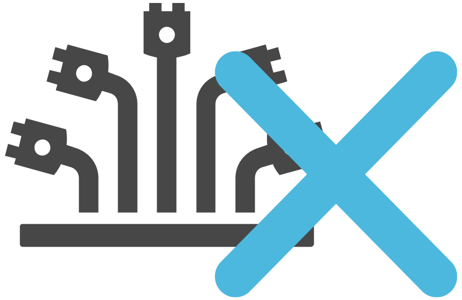
Messy wiring or multiple PCIe slots occupied?

After adopting SupremeRAID™ …

Up to 32 NVMe SSD in one Raid group

2-10x

Performance Improvement

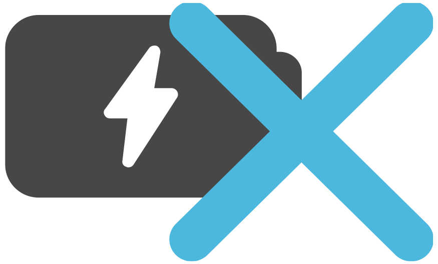


No Extra wiring

Increase signal quality

No PCIe switch

Lower cost
easy to manage



No BBU

Lower cost
easy to manage


INCEPTION PROGRAM
Graid Technology Inc.
SupremeRAID™ Leads Big Data Solutions for Smart Cities
Supreme100
powered by AMD EPYC™ 8004
100GB/s
Deliver Seamlessly
support 4-8 Gen5 NVMe drives
up to 450w mid/ high-end GPU
Edge nodes
over 2x
Performance increasing
3D Rendering
Video Editing
CAD/ Ansys
Broadcasting
Self-driving car
SIEM
ERP
Cloud Service
Smart Meter
Database
2-5x Performance
7x rebuilding time
150% more capacity
SupremeRAID™
File System
LLM
BeeGFS
lustre
LVM
SuperPOD
AI/ HPC
2-5x
Performance increasing
SupremeStore
The Cluster Storage Appliance
powered by
4th/ 5th Intel® Xeon® Scalable processor
80 GB/s
Support single chassis HA pair
5X Performance under BeeGFS
BENEFITS
from us
GPGPU RAID
4-10x
NVMe SSD Boosting
With the most secure
Data Protection
PowerEdge R760 OEM with Graid Solution

Supreme100 Workstation

POWERED BY AMD EPYC™ 8004 & SupremeRAID™ SR-1001

SPECIFICATION

Dimension: 602.6 x 175.0 x 438.0 (mm); (With bezel)
CPU: EPYC 8004 series up to 165W (optional fan solution if TDP higher than 165W)
DDR: 8 DIMM slots (1DPC/2DPC) up to 768 GB @4800 MHz
Front Panel: 1 x System reset, 1 x power on/off, 2 x USD 3.2, 1 x Audio combo
Drive Bay: 8 x 2.5“ hot-swapped NVMe SSD + 4 x 3.5” hot-swapped HDD (optional)
Expansion Slot: 2 x PCIe5.0 / CXL1.1 x16
1 x PCIe5.0 / CXL1.1 x8
2 M.2 (PCIe5.0 x4)
GPU Card: Up to 450W GPU card Double slot x 1
Rear Panel: 1 x DB15 VGA, 2 x USB 3.2, 2 x 1GbE_RJ45, 1 x MLAN, 2 x SFP28 (25GbE),
1 x UID button with LED, 1 x PWR button with LED
System Cooling: Front: 1 x 120mm fan / 2600 RPM
Rear: 1 x 80mm fan / 8200 RPM for SSD
Power Supply: ATX 1200W

Phase in Timing _ Supreme100

Meet the Requirements of High Data IO & low noise office workloads

Robust Data Protection

Equipped **4 to 8 SSDs** for the most secure data protection

Unbeatable Performance

The only workstation on the market delivers **100GB/s throughput**

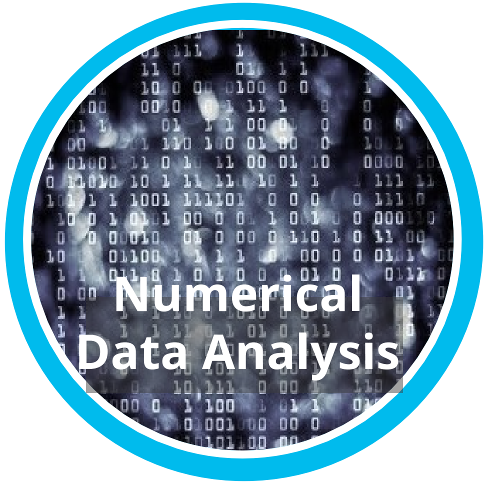
Powerful GPU support

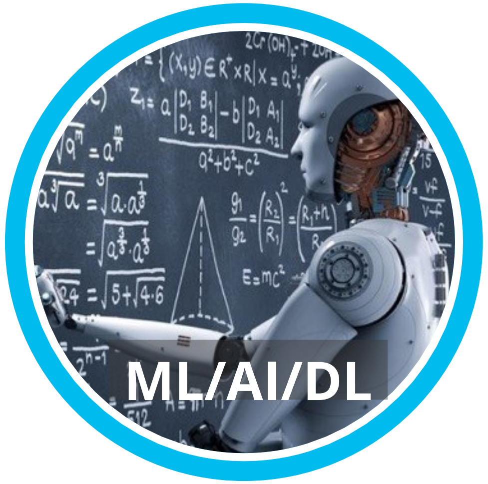
Equipped up to **450w** mid/ high-end GPU

Keep Noise Level Down

Operating noise **~ 60dB** Suitable for **indoor office**

Phsae in Territories _ Supreme100





SupremeRAID™

Solving Data IO bottleneck and saving over 2x processing time

3D Modeling

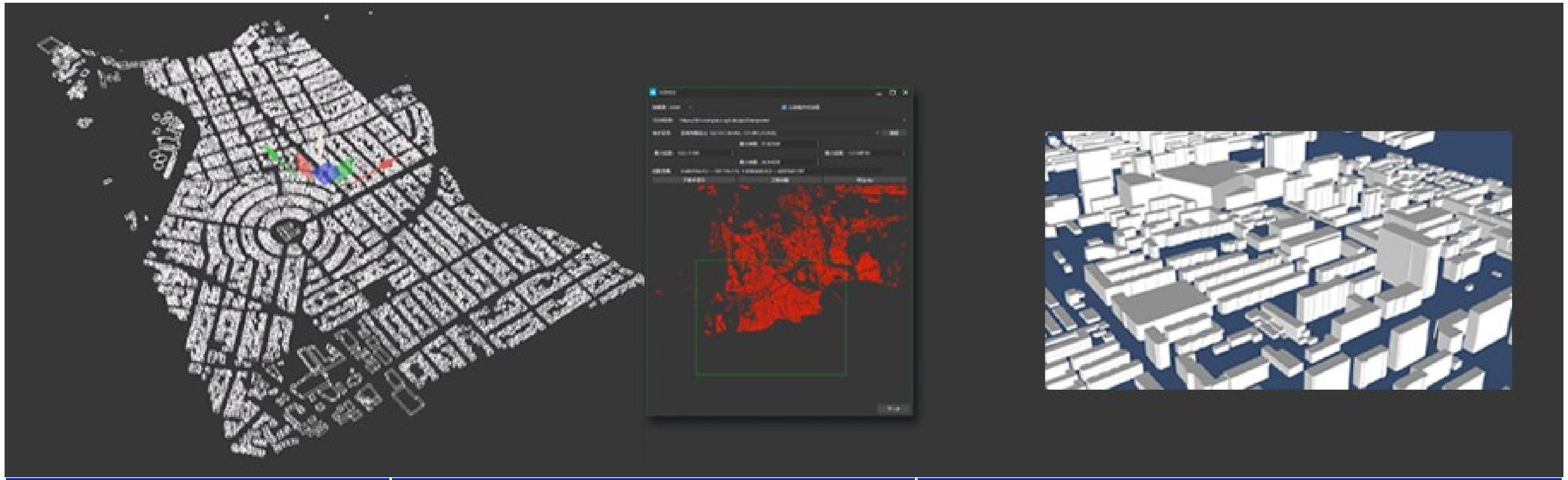
Background: Map service conducting 3D geographical modeling, using the workstation Xeon + SAS HDD RAID and SVSGeoModeler software

Pain point：
Poor performance results in a long processing time. Even increasing the CPU cannot improve overall performance

Solution： Use Intel i9 CPU +SupremeRAID™ Solution with 4 NVMe SSD to enhance 2 times performance.

Features：

- Improve the data IO performance will upgrade computing performance and save the half computing time.
- **After breaking data IO bottleneck,** upgrade CPU performance can enhance overall performance more.



	Workstation Xeon + SAS HDD RAID	Intel i9 CPU + SupremeRAID™ + 4NVMe SSD	
Free Mesh Space	3hrs 3mins	1hr 17mins	2.3x time saving
Constrained Mesh Space	1hr 16 mins	32mins	2.3x time saving
3D Model Creation	108hrs12 mins	55hrs 30mins	1.9x time saving

The Best HPC Cluster Solution Drive by SupremeRaid with BeeGFS

GRAIDTECH.COM

Storage SuperServer SSG-221E-DN2R24R

2U all-flash SBB with 24 U.2 NVMe drives and PCIe 5.0

Key Applications
Artificial Intelligence (AI), High Availability Storage, NVMe Over Fabrics Solution

Key Features
Two hot-pluggable systems (nodes) in a 2U form factor. Each node supports the following:

- Single Socket E (LGA-4677) 5th/4th Generation Intel® Xeon® Scalable processor. Up to 350W TDP.
- 2 PCIe 5.0 x16 HHHL slots
  2 PCIe 5.0 x8 HHHL slots
- Supports 8 DIMMs per node with 1DPC, up to 2TB memory capacity with 8 DIMMs of 256GB 3DS RDIMM DDR5-5600 ECC memory per node.
- Dedicated 1GbE Private Ethernet connection for node-to-node communication (Heartbeat)
- 24 Hot-swap Dual Port Gen 5 U.2 NVMe SSD bays
- Redundant Titanium 2000W Power Supplies

SSG-221E-DN2R24R

BETTER TOGETHER

Why BeeGFS + SupremeRAID™?

High Performance

SupremeStorre turnkey solution delivers extremely high performance with **80 GB/s** and lowest lantency.

Boundless Scaling out

With BeeGFS advantages, scales out storage and performance by adding more nodes, allowing for linear increases in both capacity and throughput.

Unlimited Scaling up

By utilizing SupremeRAID™, single machine provides ultra-high throughput and easily scaling up with increasing data demands.

Body Mirror by BeeGFS and Cross Node HA by SupremeRaid

Provides high availability, ensuring data access and business continuity even during partial system failures.

Adapt to Diverse Workloads

Whether facing high-density data processing tasks or random I/O operations, our system ensures efficient performance.

Native CSI Driver

Supports the latest CSI specifications for seamless integration with modern container infrastructure, ensuring flexible deployment and scalability.

Phase in Timing _ SupremeStore

Best choice for Small/ Mid-Sized AI Data Center
High Performance and Scalability

BeeGFS®

*Taiwan Certified Reseller

Outstanding Performance

up to **80GB/s** with one node, support for 32 HGX.

Performance can be **linearly scaled,** from TB to PB level, easily configured according to budget.

High Scalability

With SupremRAID™, flexible expansion up to **20 SSDs**, enabling effortless linear performance growth.

With BeeGFS horizontal scaling, capacity can be expanded to over 1000 nodes.

Security Protection

Dual-controller devices paired with SupremeRAID™ **high availability (HA) technology**.

BeeGFS **Body Mirror technology** to provide enhanced data protection.

Reliable Service

5x8 NBD and **7x24x4** on-site repair services and remote technical support

Graid is the authorized reseller in Taiwan, offering professional BeeGFS training.

Phsae in Territories _ SupremeStore

The best choice of AI/ HPC & Video Editing

SuperPOD
AI/ ML

Video
Editing

HPC
academic research

AI SuperPOD: The best Turnkey solution team

Data Scientist/
Developer/
AI Researcher

Generative AI

INFINITIX

AI-Stack

Self- Service, Scalable Infrastructure	Data Science& AI Workload Orchestration	Reproducibility & Collaboration	Governed Utilization of GPU Resource	Infrastructure Optimization	Cloud-Native Deployment	AI and Data Science Tolls and Frameworks

Container Orchestration and Management Integration

kubernetes

IT Administrator

Mainstream GPU Accelerated Servers

NVIDIA HGX

SUPERMICRO

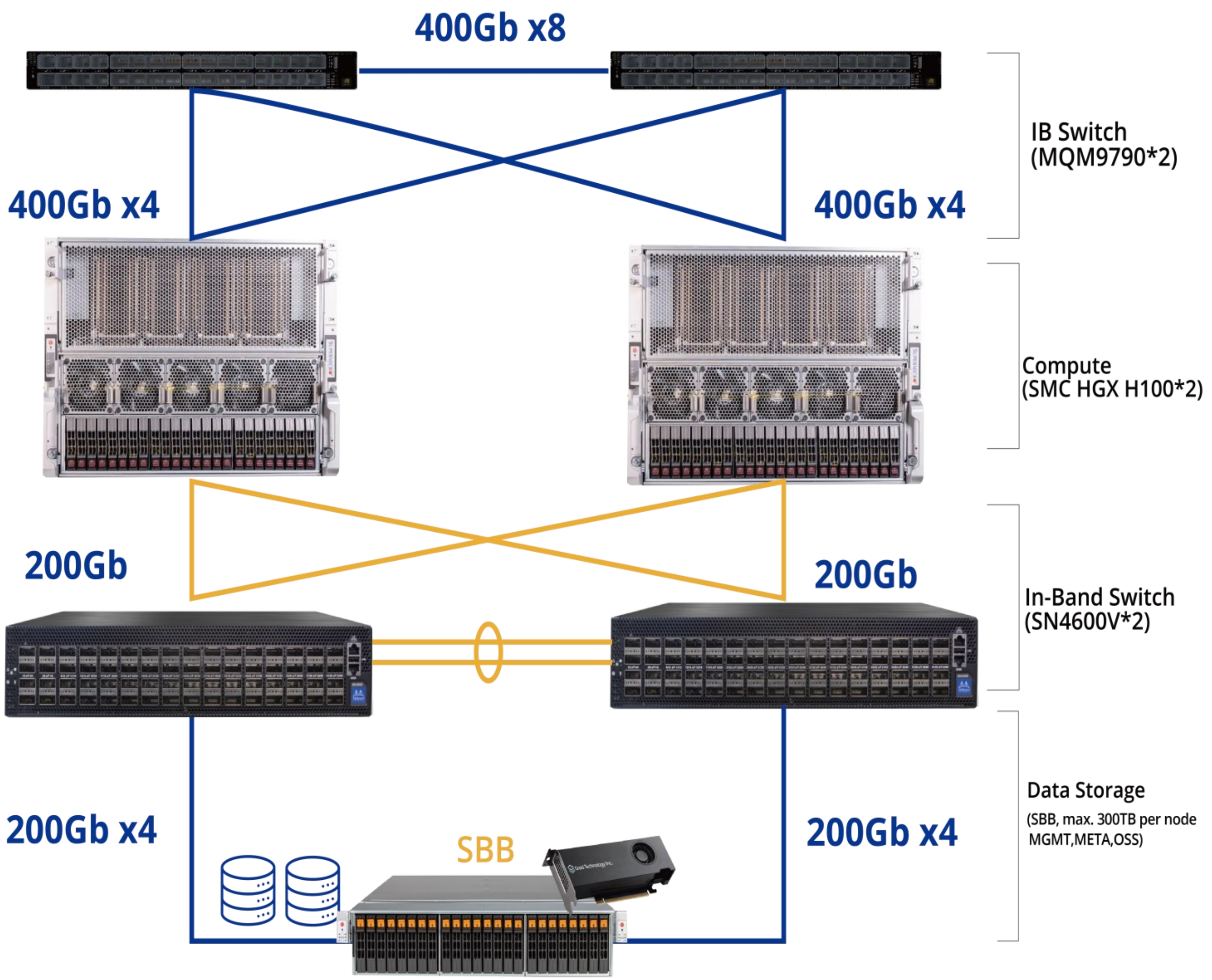
Graid Technology Inc.

Accelerated Storages

BeeGFS

MINIO

SuperPOD Architecture



SupremeStore

Turn your SuperPOD into a real business opportunity.

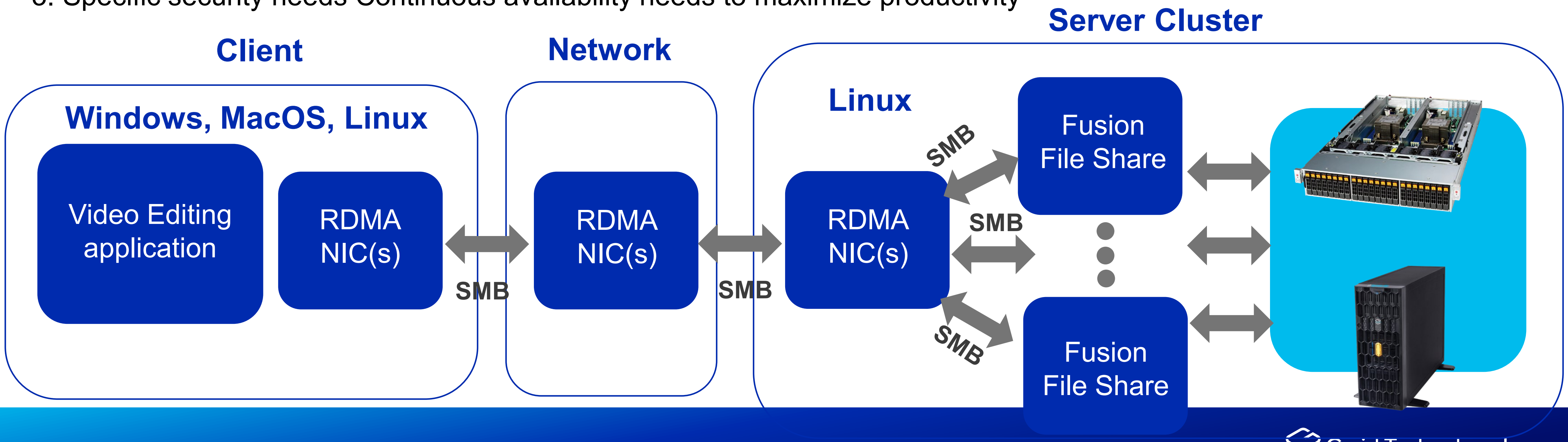
Video Editing

SupremeStore x TUXERA

60 frames per second real-time edit uncompressed 4k videos from share file cluster

Pain Point：

1. Transfer rate needed for 8k uncompressed video with 60 fps is ~16.2GB/s, Open-source SMB solution provides max 2-3 GB/s transfer speed
2. Fast storage not in full usage.
3. Specific security needs Continuous availability needs to maximize productivity



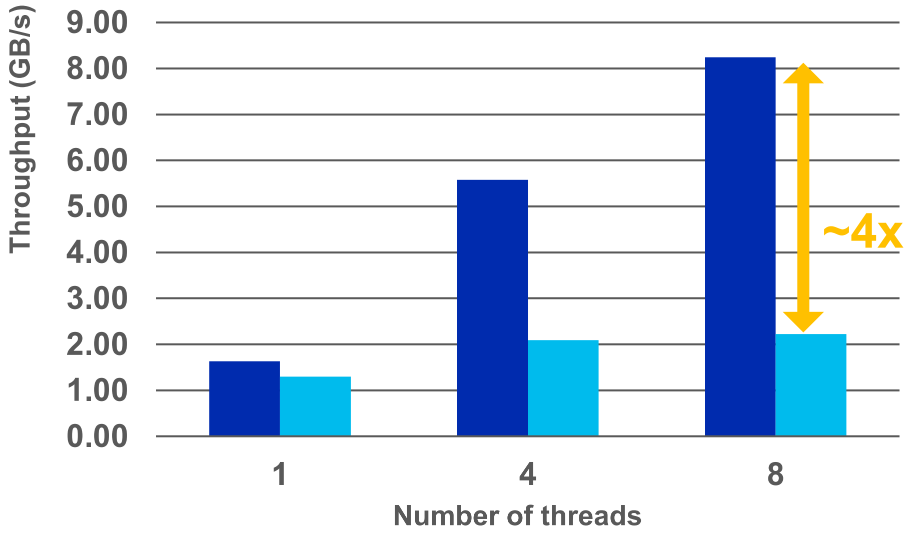
Video Editing

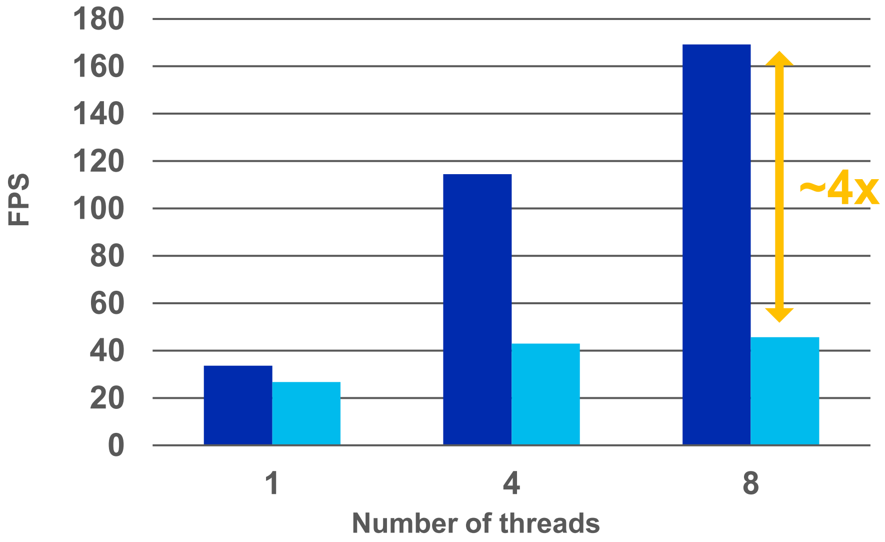
SupremeStore x TUXERA

Up to 4 times throughput/ FPS performance

Sucessful connection , connecting 200 clients per second

workload performance comparison

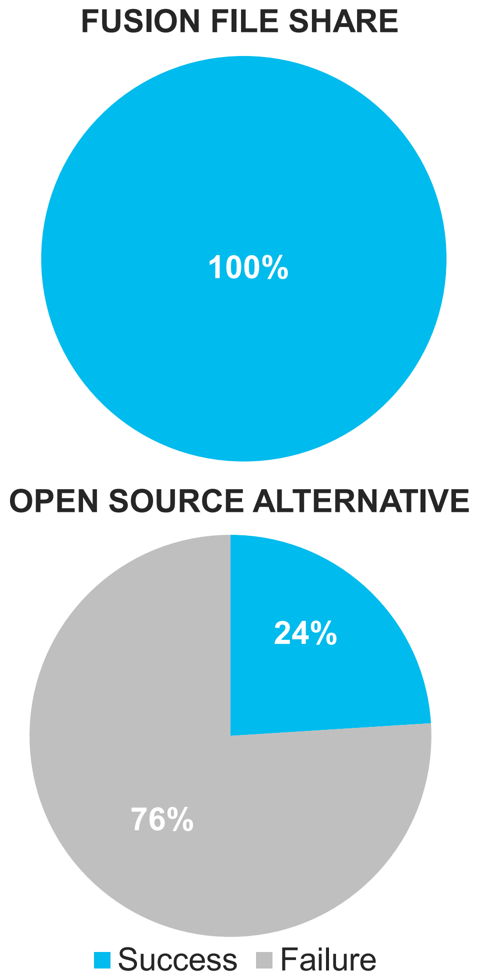




Higher values are better.

■ Fusion + SupremeStore ■ Open source alternative

SMB Connection Rate



Dell PowerEdge R760 OEM with Graid Solution

2U all-flash Server with 24 Gen4 U.2 NVMe drives

Key Applications
Designed to deliver not only peak compute performance but also best Data IO with SupremeRaid. Optimizes the most demanding workloads like
Artificial Intelligence (AI), HPC, Database, Video editing, 3D rendering

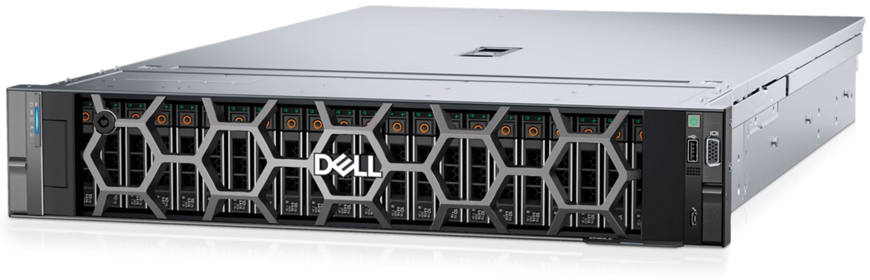
Key Features

- 2.5” chassis up to 24 NVMe Direct Drives,
- 2 CPU:

CPU: Intel® Xeon® Gold 6430 2.1G, 32C/64T, 16GT/s, 60M Cache, Turbo, HT (270W) DDR5-4400 x 2

- Memory: 16GB x 12 = 192GB
- HDD: BOSS-N1 controller card + with 2 M.2 480GB (RAID 1)
- Broadcom 5720 Dual Port 1GbE LOM
- Broadcom 57414 Dual Port 10/25GbE SFP28, OCP NIC 3.0 with 10G transceivers

NO OS
Prosupport NBD



PowerEdge R760 powered by Graid

SupremeRAID™ Global Partners

HPC/AI

Aspen Systems Inc.
MAZDA COMPUTING
CRYSTAL
Agency for Defense Development
HPCLAB
MDSTECH
ADVANCED HPC
DATA IN SCIENCE TECHNOLOGIES DST
QuoPin
GPL TECHNOLOGIES
A-VAR
HPC SYSTEMS
SHANGHAITECH UNIVERSITY
EXXACT
TRENTON SYSTEMS
ETRI Electronics and Telecommunications Research Institute
Lunit
大綜電腦系統 JETWELL
LIQID
TOKAI UNIVERSITY
VUNO View the Invisible, Know the Unknown
nextron
genesis lab
sysGen
NARIT
MiruWare System & Solution
define tech
PEKING UNIVERSITY 1898
华大智造 MGI
INFINITIX

*The list of collaborating partners is extensive; only representative examples are provided.

SupremeRAID™ Global Partners & OEMs

Storage/file system

Database

*The list of collaborating partners is extensive; only representative examples are provided.

SupremeRAID™ Global Partners & OEMs

Packet capture/ cybersecurity

Media Broadcasting

3D Rendering/ Gaming

Graid Technology Inc.

The Safest SSD RAID Card

1. **Support RAID Levels:** 0/ 1/ 5/ 6/ 10
2. **Controller Failover HA & Cross node HA**
3. **Double Failure Protection with Distributed Journaling:** 2nd Journal in SSDs when RAID Degrade
4. **Coming soon in 2025:**
   Non-disruptive rebuild (Data backed up to a new SSD before failure), and error retry mechanism on PD, Erasure code

Brilliant Figure with SupremeRAID™

Since 2021 Graid named over 10 famous awards in worldwide

APICTA Asia Pacific ICT Alliance AWARDS

2023 APICTA

APICTA Named **Golden Award** in BS category. Judges were especially impressed with its ability to maximize NVMe SSD performance in AI and high-performance化 NVMe SSD

- Hong Kong

FMS

2024 Flash Memory Summit

Joint solutions with partners won **the Most Innovative Enterprise Business Application** and **Best of Show awards** for excellence in high-performance media and entertainment applications.

- The U.S.

SupremeRAID™ 2025 Driver Release Roadmap

SR Windows Driver 1.2.4

- Adds support for the management **GUI.**
- Adds support for SR-1000-AM **(NVIDIA RTX A1000) ➔phase in 2026**

SR Linux Driver 1.7

- Adds **error retry mechanism** on physical drives to **enhance drive group robustness**.
- **Provides drive copyback functionality** to mitigate performance impact and **reduce risk during drive replacement.**

SR Linux Driver 2.0

- Improves RAID5/6 4K random write IOPS. **➔ 2~3 X**

SR Windows Driver 1.2.5

- Minor enhancements and bug fixing.

SR Linux Driver 2.1

- Supports up **to 64 PDs**.
- Adds support for drive group expansion.
- Implements erasure coding.

SR Windows Driver 1.6

- Aligns with Linux Driver 1.6 features.

2025 Q1 — 2025 Q2 — 2025 Q3 — 2025 Q4 — 2026 Q1

SupremeRAID™ SE Beta

- Enables RAID5 with 4 to 8 drives running on the customer's own GPU.
- Introduces subscription licensing model and the central user portal.

SupremeRAID™ AE Release

- Supports GPUDirect storage.
- Intelligent data offload engine.
- Integrates with parallel filesystems and SDS GPU server convergence deployment.

SupremeRAID™ HE Release

- Supports cross-node HA capability.
- Supports failover mechanism for BeeGFS and SMC SBB systems.

SupremeRAID™ SE Release

- Enable subscription payment flow.

New Product Lines

A powerful engine, various scenarios

Enterprise / Datacenter

Individual

SupremeRAID™ SR

- Original product line, general purpose NVMe RAID for enterprise and data center.
- Linux & Windows

SupremeRAID™ AE (AI Edition)

- Tailored for GPU servers and AI workloads. Featuring GPUDirect Storage support and an Intelligent Data Offload Engine.
- Linux Only.

SupremeRAID™ HE (HPC Edition)

- Offers cross-node HA capability, enabling seamless integration with diverse parallel filesystems and SDS solutions to deliver a streamlined and high-performance storage solution.
- Linux Only.

SupremeRAID™ SE (Simple Edition)

- Provides enterprise-grade RAID data protection and unparalleled NVMe RAID performance at the desktop level.
- Linux & Windows

Please Contsct

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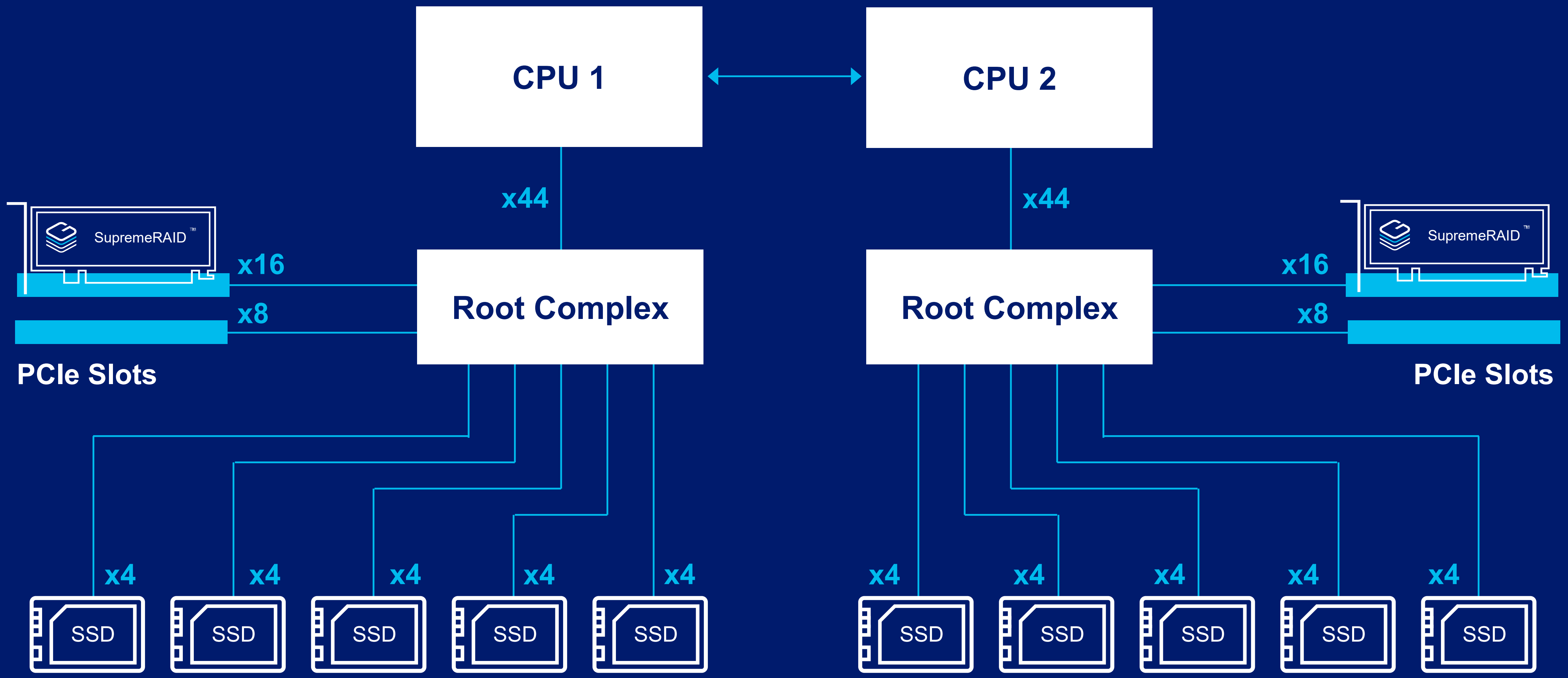
Thank you

How SupremeRAID™ Fits: High Available

Active-Active & Active-Standby



Why the Double Failure Matters

How Does RAID Protect Data in a Failure Event?

- Drive failure (RAID degradation) -> RAID Redundancy
- System abnormal shutdown -> Re-synchronize the stripe

What is the double failure?

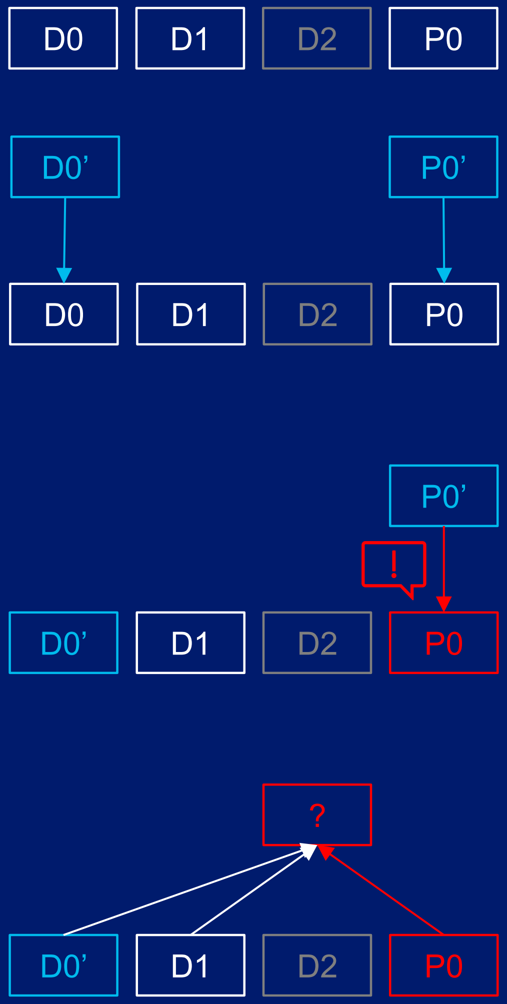
- **RAID Degradation + System Abnormal Shutdown**

How it occurs?

- The RAID group has degraded.
- Write Operation: Data and parity are being written to the disks.
- Unexpected Shutdown: Power loss or system crash interrupts the write process.

Result:

- Completed writes to some disks but not others, causing mismatched data and parity. **This mismatch will cause the missing chunk corrupted when performing the degraded read.**



A degraded RAID5 Stripe, where D is the data, and P is the parity P0 = D0 ⊕ D1 ⊕ D2. The D2 drive is missing.

When updating D0 to D0', P0 must also be updated to P0', where P0' = D0' ⊕ D1 ⊕ D2.

A system crash occurred during the operation: D0 was updated to D0', but P0 wasn't, making the stripe inconsistent because P0 ≠ D0' ⊕ D1 ⊕ D2.

If the application reads D2, a degraded read will be performed, but D0' ⊕ D1 ⊕ P0 no longer equals D2.

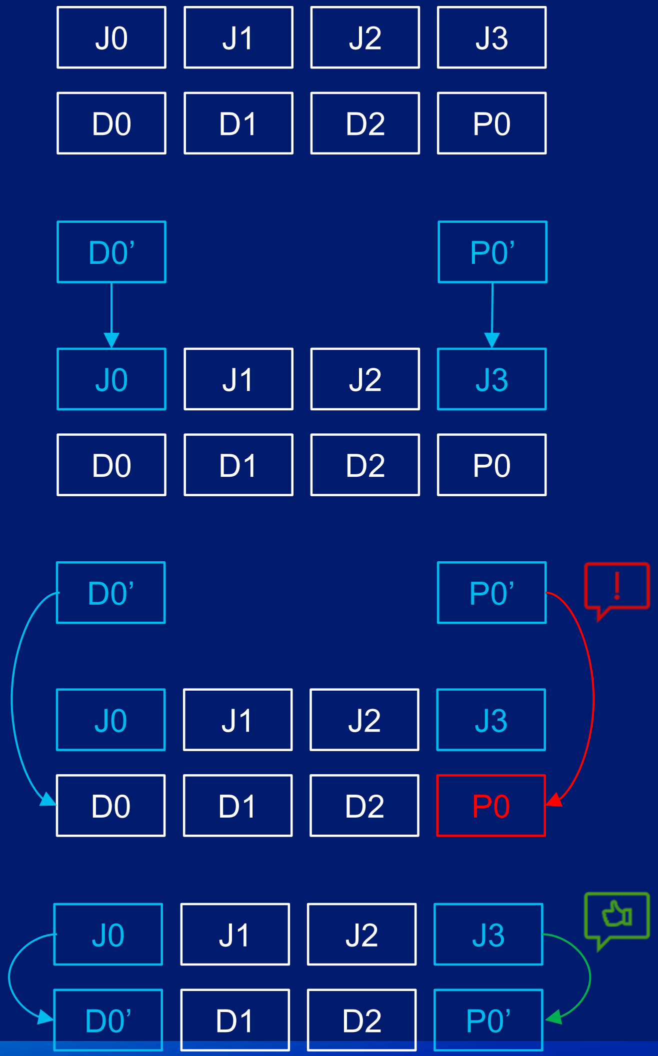
Write Hole Protection with Distributed Journaling

A RAID5 Stripe, where D is the data, and P is the parity
P0 = D0 ⊕ D1 ⊕ D2

When updating D0 to D0', and P0 to P0', the updates for D0' and P0' are first stored in J0 and J3

Once D0' and P0' have been stored in J0 and J3, the write operation to D0 and P0 will be executed. If a system crash occurs during the update of D0 and P0, resulting in P0 not being updated

After the system comes back online, we can use J0 and J3 to ensure that both D0 and P0 have been updated to D0' and P0', respectively.



What is Journaling?

Journaling involves storing data temporarily in a designated area before executing the actual write operation. This process ensures that, in the event of a system crash during writing, the journal can be used to repair any incomplete data.

Why Distributed?

Distributed journaling ensures performance is not limited by a single drive while also preserving redundancy, distributing the journaling process across multiple drives to enhance both speed and reliability.